

VARIA Carbon – AI-Powered Soil Carbon MRV

The Problem

Lab-based soil carbon measurement costs \$30-50 per hectare and does not scale. For smallholder farmers and landscape-level projects, this makes MRV economically unviable.

The Solution

VARIA Carbon predicts Soil Organic Carbon (SOC%) from satellite data — no lab, no field visit.

Inputs (5 variables):

- NDVI (from satellite imagery)
- Annual precipitation (mm/year)
- Mean annual temperature (°C)
- Elevation (m)
- Clay content (%)

Output: SOC% + carbon credits potential (tons CO₂ per hectare)

Technical Performance:

- $R^2 = 0.78$ (validated on EU LUCAS soil database)

- MAE = 0.80% SOC
- Trained on 70 real soil samples
- No overfitting (train/test R^2 difference <0.15)

Why Project Developers Choose VARIA Carbon:

- Affordable: Lab tests cost \$30-50/ha — VARIA Carbon offers a one-time license with unlimited use
- Scalable: Works across thousands of hectares in seconds
- Transparent: Full code included (no black box)
- Private: Deploys on your own server (data never leaves your organization)
- Easy: One-command Docker installation, interactive dashboard, and REST API

Use Cases for Carbon Projects:

- Baseline soil carbon assessment
- Cost-effective MRV at scale
- Prioritization of high-potential areas
- Farmer-level data systems

Pricing (One-Time, Perpetual License):

Academic / Research: \$499 (Full code + model + 1 year updates)

Commercial / Business: \$1,999 (Full code + model + priority support)

Enterprise (IP Transfer): Contact for pricing (Complete IP + training + 5 years support)

Delivery:

- Digital download (2GB zip file)
- Docker installation guide (5 minutes)
- 14-day money-back guarantee

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Ready to deploy. Ready to scale.